

The active nonsmooth manifolds of a neural network classifier: a tool for confidence assessment

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Abstract. This paper proposes a new approach to the study of the important question of assessing the reliability of a deep network in a classification. Whereas many previous approaches make use of random perturbations, our approach, based on the concept of active manifold, builds a trajectory from any new data point, or any point in the training set as well, which follows the steepest descent curve associated with the softmax components indexed by its label. Along this trajectory, a tipping point can be defined as the first point at which the predicted label switches to a different one. The distance of each sample point to its associated tipping point along its active manifold provides the user with a robustness index, which can be compared across all points in a reference set in order to probe the data’s distribution and discover the specific sub-areas of its support where classification cannot be made with large confidence. As compared with randomised approaches, our method is entirely driven by the network itself without the need of prior statistical information. In addition, we also empirically show that our approach provides a simple and efficient two stage approach to data augmentation by suggesting to incorporate in the training set the nearest tipping points discovered in the first stage.

1 Introduction

Since the early days of Deep Learning research, deep neural networks have been extensively tested on the basic problem of text, image or signal classification. Every month, a novel architecture is being proposed in the literature with always better performance. This constant increase of efficiency has kept driving an extensive interest from researchers in academia and the industry as well. Applications in real life are innumerable in health, manufacturing, entertainment, etc.

Current neural network architectures are rarely underparametrised. It even came as a big surprise that highly overparametrised networks often show impressive generalisation performance in many settings, a very counterintuitive property from a purely statistical perspective. Still today, the actual impact of

overparametrisation is one of the most intriguing open questions in deep learning. Breakthrough work of Belkin and co-authors recently showed that least norm solutions to empirical risk minimisation can generalise, and as a result, they brought to the community’s attention what is now best known under the name of “the double descent phenomenon”. Another approach, based on the remark that the optimisation algorithm used in the training phase can play a role in the performance of the trained solution, is also greatly investigated and a property called “implicit bias” has been discovered and proved to implicitly penalise the empirical risk, and thus reduce at the same time the complexity of the family of candidate networks, left aside the problem of appropriately tuning the hyperparameters, such as the learning rate and the batch size. Neural Tangent Kernels (NTK) also provide an intriguing viewpoint on controlling the theoretical generalisation error in certain situations where the training algorithm converges in the vicinity of the random initial point, hence allowing a Taylor expansion to linearise the analysis [4].

Nevertheless, despite these many evidence of practical successes and theoretical advances, numerous open questions remain, and in particular the one of assessing the robustness of a network and the confidence that can be drawn from the obtained predictions on new data points. Discovering tools that can unveil the lack of reliability of a given network at hand is of urgent need for practitioners which need to avoid the pitfall of overconfidence in neural network technology. This is even more critical when the training sets in some applications are smaller than the training sets on which the successful networks were showcased in the literature or the media. In poorly trained settings, which is the case in a vast majority of industrial and research applications, overparametrised networks may dangerously overfit. The case of clever Hans is an illustration that what the network actually learns about the data may be far from the features that a rigorous understanding would suggest in order to make a sound decision about any new input to classify [7]. Closely related is the search for counterfactuals, i.e. “how the world would have (had) to be different for a desirable outcome to occur” [10]. Finding nearest counterfactual explanations, i.e. instances which are close in some metric to the well classified inputs is an idea that made its way to the current trend in XAI [8,5]. It is also equally well known that image classifiers can show embarrassing weaknesses in the face of small adversarial perturbations [9], whose existence is a warning about their potential unreliability.

We address the problem of quantifying the robustness of a neural network classifier by proposing a principled approach to data perturbation which applies individually to both training data or new data as well, via the building of a steepest descent trajectory for the softmax component of the initial predicted label. This trajectory is also known as Active Manifold in the smooth setting [1] and is an extension of the celebrated method of Active Spaces in Uncertainty Quantification (UQ) [2]. Our contributions are twofold: first we show that our method is able to quickly discover nearby perturbations of any new data point or any point in the training set as well, for which the label is different from the initial point, a sort of counterfactual near the training set which gives a

natural way of probing the dataset distribution for unveiling its potential lack of reliability. Second, our method automatically suggests new data points for augmenting the training set, as the discovered tipping points which are “too near to have a different label”, with possible input from human experts in the loop.

1.1 Background on deep learning and active manifolds

1.2 The neural network model

Assume that all the data have the same size, say $d_1 \times d_2 = d > 0$ with real pixel values. A classifier is a mapping

$$f : \mathbb{R}^d \rightarrow \mathcal{Y}$$

with $\mathcal{Y}$ a finite set of C labels being identified with $\{1, \dots, C\}$. In the sequel, we assume that the classification f is a deep neural network architecture. At a point where f is differentiable, we have

$$f(x) = \sigma_{out}(\mathcal{A}_L(\sigma(\mathcal{A}_{L-1}(\dots \sigma(\mathcal{A}_1(x))))), \quad (1)$$

where σ is a non-linear function such as the sigmoid or the RELU, and the affine operators $\mathcal{A}_l$, $l = 1, \dots, L$ are the parameters which have been trained using a training dataset, and the last layer is defined by selecting the usual softmax output, i.e. it has the form

$$\sigma_{out}(z)_c = \frac{e^{z_c}}{\sum_{c'=1}^C e^{z_{c'}}}$$

for $c = 1, \dots, C$ and $z = (z_1, \dots, z_K) \in \mathbb{R}^C$. The predicted class is chosen according to the rule

$$\hat{c}_x \in \operatorname{argmax}_{c \in \{1, \dots, C\}} f_c(x). \quad (2)$$

1.3 Active manifolds

Let x_i be a data point from the training sample. Let $\hat{c}_i := \hat{c}_{x_i}$ denote the corresponding index given by (2). We now will devote some work on understanding the function $f_{\hat{c}_i} : \mathbb{R}^d \mapsto \mathbb{R}$, i.e. the $\hat{c}_i$ -th coordinate of f , and the associated “active” manifold.

Recalls on active spaces Principal Component Analysis is the study of the main intrinsic linear features of a dataset. In the same spirit, active space theory tackles the problem of finding a subspaces of low dimensionality which best captures the variations of a given function $\phi : \mathbb{R}^d \mapsto \mathbb{R}$, hence capturing the intrinsic dimensionality of ϕ . The notion of active space was demonstrated very powerful and has had a substantial impact on sensitivity analysis and model reduction in many applications [2].

A non-linear analog: active manifolds Our work will use a nonlinear counterpart to active spaces, namely active manifolds [1]. The first one-dimensional active manifold is a curve which captures the best one-dimensional nonlinear subdomain where the function has the largest amount of variations. All the information about the studied function is captured on a one dimensional manifold. The active manifold, associated to a $\mathcal{C}^1$ function $f: \mathbb{R}^d \mapsto \mathbb{R}$ is a differentiable curve satisfying the differential equation

$$\gamma'_x(t) = -\frac{\nabla f(\gamma_x(t))}{\|\nabla f(\gamma_x(t))\|} \quad (3)$$

which is a well defined ODE so long as $\|\nabla f(\gamma_x(t))\| \neq 0$. The path $t \mapsto \gamma_x(t)$, when well defined, is called the active manifold passing through x . Computation of the active manifold is usually achieved using Euler’s discretisation method.

2 The nonsmooth active manifold of a deep neural network

Let us now consider how active manifolds can help us follow a fast path to a nearby counterfactuals [6] on the active manifold. We start with the preliminary question of extending the notion of active manifold to the case of differential inclusions which is the relevant notion in the case of deep neural networks with ReLU activation functions. For any $x_i \in \mathbb{R}^d$ with predicted label $\hat{c}_i = \hat{c}_{x_i}$. Then, assuming we can define an active manifold γ_{x_i} for ReLU activated networks, one can expect to reach a tipping point $\hat{x}_i$, at which the network’s decision changes labels for the first time. The distance from the original image x_i to the tipping point $\hat{x}_i$ will help define a confidence metric for the classifier under study.

2.1 Recalls on differential inclusions

Let $I = [0, T]$ be a finite interval, $x \in \mathbb{R}^n$, and F be a map from $I \times \mathbb{R}^n$ into the set of all subsets of $\mathbb{R}^n$. Find an absolutely continuous function $\gamma_x(\cdot)$ on I such that $\gamma_x(0) = x$ and

$$\dot{\gamma}_x(t) \in F(t, \gamma_x(t)) \quad (4)$$

for almost all $t \in I$ where $\dot{\gamma}_x(\cdot)$ is the derivative of $\gamma_x(\cdot)$.

Let Γ be the set of solutions to (4). As a rule, the set Γ consists of more than one element, that is we have a bundle of trajectories. Consequently, there are various closely connected approaches of approximating solutions $\gamma_x(\cdot) \in \Gamma$.

2.2 Constructing an approximate curve on the nonsmooth manifold

Let $\eta_0 = \gamma_x(0)$, and for $j = 0, \dots, N-1$ compute η_{j+1} using the iterations

$$\eta_{j+1} \in \eta_j + hF(t_j, \eta_j) \quad (5)$$

on a uniform grid

$$0 = t_0^N < t_1^N < \dots < t_N^N = T \quad (6)$$

with constant stepsize h , i.e.

$$h = t_j^N - t_{j-1}^N, \quad (7)$$

for $j = 1, \dots, N$. Consider the piecewise linear interpolation of this discrete sequence, i.e.

$$\eta^N(\cdot) : I \rightarrow \mathbb{R}^n \quad (8)$$

such that

$$\eta^N(t) = \eta_j + \frac{1}{h} (t - t_j) (\eta_{j+1} - \eta_j) \quad (9)$$

for $t_j \leq t \leq t_{j+1}$, $j = 0, \dots, N-1$.

The following theorem, nicely presented in [3], gives easy to verify conditions under which there exists a subsequence that converges to a solution of (4).

Theorem 1. *Suppose that the set-valued map F satisfies the conditions:*

1. *F is nonempty compact and convex valued upper semicontinuous in $I \times \mathbb{R}^n$.*
2. *There exist constants k and a , such that*

$$\|z\|_2 \leq k\|x\|_2 + a \quad (10)$$

whenever $z \in F(t, x)$, $x \in \mathbb{R}^n$, $t \in I$.

Then every sequence $(\eta^N(\cdot))_{N \in \mathbb{N}}$ defined by (9) for $N \in \mathbb{N}$ has a subsequence which converges as $N \rightarrow \infty$, uniformly in I , to some solution of the problem (4).

2.3 The Nonsmooth active manifold of a ReLU networks as differential inclusions

The main tool of our paper is a generalisation of the notion of active manifold to the nonsmooth case and its relevance for the study of deep ReLU networks.

Definition 1. *The nonsmooth active manifold, associated to subdifferentiable function $f: \mathbb{R}^d \mapsto \mathbb{R}$ is the set of solutions to the differential inclusion*

$$\gamma'_x(t) \in -\eta_t \partial f(\gamma_x(t)), \quad (11)$$

where η_t is chosen so that $\|\gamma'_x(t)\|_2 = 1$.

We now have to check that the *non-smooth active manifold* of any component of the output function of a ReLU network satisfies the conditions of Theorem 1 and we are interested in the component indexed by c of this function, i.e. f_c . Recall that in such a case, the function f has the form (1). Let us define

$$f^{(l)}(x) = \mathcal{A}_l(\sigma(x)),$$

The gradient at a point where f_c is differentiable is given by

$$\begin{aligned} \nabla_x(f_c)(\mathcal{A}; x) &= \left(J_x f^{(1)} \right)_x^\top \left(J_{f^{(1)}(x)} f^{(2)} \right)_{f^{(1)}(x)}^\top \cdots \\ &\quad \left(J_{f^{(L-1)}(x)} f^{(L)} \right)_{f^{(L-1)}(x)}^\top \nabla \sigma_{out}(f^{(L)}(x))_c \end{aligned}$$

where $(J_{f^{(l-1)}} f^{(l)})_{f^{(l-1)}(x)}$ denotes the Jacobian of $f^{(l)}$ along the variable $f^{(l-1)}(x)$ and evaluated at $f^{(l-1)}(x)$. Now since all the partial derivatives involved are bounded by a constant, inequality (10) is satisfied and the convergence of the Euler approximation is guaranteed.

2.4 Tipping points and the stability metric

Let x be an image classified as $\hat{c}_x \in \{1, \dots, C\}$ by the neural net. A tipping point for the neural net f associated with the sample x is first point $\tilde{x} \in \mathbb{R}^d$ which is obtained on an active manifold path, and such that f stops classifying as $\hat{c}_x$. Mathematically speaking, given a trajectory γ_x on the nonsmooth manifold associated with x , a tipping point $\tilde{x}$ for f associated with the sample x is defined as:

$$\tilde{x} = \gamma_x(t_{tip})$$

where

$$t_{tip} := \inf \left\{ t \in [0, 1], f_{\hat{c}}(\gamma_x(t)) < \max_{1 \leq c \leq C} f_c(\gamma_x(t)) \right\}.$$

Finally we define the *confidence index* of f at the image x by:

$$\mathcal{M}(f, x) = \int_0^w \|\gamma'_x(t)\| dt. \quad (12)$$

In other words, the safety distance $\mathcal{M}(f, x)$ is the length of the active path that connects the original image x to its tipping point $\tilde{x}$ where a different label is predicted for the first time along the active curve.

3 Multi-active manifolds: accounting for all the classes

In this section, we extend the idea of nonsmooth manifold of a network in order to account for all the components of the softmax function. As a result, we obtain

a trajectory which reaches a potentially closer and more meaningful tipping point than by using the nonsmooth manifold from the previous section. Moreover, the method of the previous section is a special case of the new scheme presented hereunder.

Let x be an image classified by the neural net as belonging to the class labeled by $\hat{c} \in \{1, \dots, C\}$. We consider for $\hat{c}$ and any $c \in \{1, \dots, C\}$ with $c \neq \hat{c}$ arbitrary solutions of the following two problems

$$\begin{aligned} \gamma_x^{\hat{c}} &= \operatorname{argmin}_{\substack{\gamma(0)=x \\ \|\gamma'\|_2=1 \text{ a.e.}}} \int_0^1 \left\langle \nabla f_{c'}(\gamma(t)), \frac{d\gamma}{dt}(t) \right\rangle dt \\ &\text{and} \\ \gamma_x^c &= \operatorname{argmax}_{\substack{\gamma(0)=x \\ \|\gamma'\|_2=1 \text{ a.e.}}} \int_0^1 \left\langle \nabla f_c(\gamma(t)), \frac{d\gamma}{dt}(t) \right\rangle dt. \end{aligned}$$

An easy computation gives

$$\gamma_x^{\hat{c}}(t)' \in -\eta_t \partial f(\gamma_x^{\hat{c}}(t)) \quad (13)$$

which recovers the previous nonsmooth active manifold of $f_{\hat{c}}$. Notice further that for all $c \neq \hat{c}$, we have

$$\gamma_x^c(t)' \in \eta_t \partial f(\gamma_x^c(t)), \quad (14)$$

where η_t is chosen so that $\|\gamma_x^c(t)'\|_2 = 1$.

Notice in particular that we are now looking for trajectories that maximise the ascent along the direction of the gradient of the concurrent components of the softmax function.

Our generalised nonsmooth manifold of the network at x is defined as the following convex combinations of the trajectories γ_x^c , $c \in \{1, \dots, C\}$ defined above.

$$\gamma(t) = \sum_{c=1}^C \mu(t)_c \cdot \gamma_x^c(t) \quad (15)$$

with

$$\mu : [0, \infty[\rightarrow [0, 1]^C$$

and $\sum_{c=1}^C \mu(t)_c = 1$ for all $t \in [0, \infty[$. We will focus on the following different choices for μ :

- “SoftMax”, corresponds to $\mu = \text{SoftMax output of the neural net}$.
- “Dirac”, corresponds to a $\mu_{\hat{c}} = 1$ and $\mu_c = 0$ for $c \neq \hat{c}$.
- “ $\frac{1}{2}$ second max and $\frac{1}{2}$ max”, corresponds to $\mu_{\hat{c}}(x) = \frac{1}{2}$, $\mu_{\hat{c}^{(2)}}(x) = \frac{1}{2}$ and $\mu_c(x) = 0$ for $c \notin \{\hat{c}, \hat{c}^{(2)}\}$, where $\hat{c}^{(2)}$ is the indice of the second maximum of the softmax function at the sample point x .

- “Shifted output vector”, corresponds to

$$\mu(x) = \frac{\hat{f}(x) - \alpha(x) \cdot (1, \dots, 1)}{\|\hat{f}(x) - \alpha(x) \cdot (1, \dots, 1)\|}, \quad (16)$$

with $\hat{f}$ the same neural net as f but without the softmax layer, and

$$\alpha(x) = \min_{1 \leq c \leq C} \hat{f}_c(X) \quad (17)$$

the minimum value of the output vector $\hat{f}(x)$.

We will study the respective performance of these different implementations in Section 4.

4 Confidence analysis, unsupervised feature delineation and data augmentation via active manifolds: one stone for three birds from very different species

In this section, we study three different challenges for which using the active manifold approach can provide an efficient approach: confidence analysis via the confidence index, unsupervised feature delineation and automatic data augmentation. We perform a series of numerical experiments that illustrate the various properties of the different proposed configurations of the active non-smooth manifolds with respect to these tasks.³

4.1 Computational speed of solving the active manifold problem for each setup

Studying the computational effectiveness of a method is a first criterion of interest in any machine learning methodology. The table below summarises the computational time for each active manifold path under study. These simulation results are displayed in figures 1, 2, 3, 4 below.

Name	Time
SoftMax	22min 47sec
Dirac	37min 38sec
$\frac{1}{2}\max + \frac{1}{2}\text{second max}$	22min 1sec
Shifted output vector	33min 4sec

We see from these experiments that the method provides a solution to the problem of computing the active manifold and the associated tipping point in quite a short computational time.

³ All our experiments are based on fine tuning of ResNet50, ResNet101 and ResNet152 for image classification tasks. We used the Tiny ImageNet dataset available on Kaggle, which consists of 500 images per class. We selected 7 image classes from the initial dataset, and then randomly removed 50 images from each class for the purpose of building the balanced validation data set.

4.2 The confidence index : comparing the different non-smooth active manifolds

Our first numerical results illustrate how the method of active manifold can be used to analyse the confidence we can have into the label assigned by the network to a given image. Figures 1, 2, 3, 4 provide an boxplot based visualisation of the distribution of the confidence index (12) for each image class: the higher the results the more confidence we can have about the network’s decision.

Arguably the best choice in this context was the path “ $\frac{1}{2}$ max + $\frac{1}{2}$ second max”. Indeed this path: 1) is statistically the least time costly 2) in comparison with the other paths 1, 2, 3, 4 allows to find the closest tipping points. The tipping path “SoftMax” also has a similar behaviour. This similarity is probably due to the fact that all the classes below the second best are squashed by the softmax function. The subsequent stability studies and the Euclidean distance studies of the deeper ResNets are done with a fixed step of 0.01 by using the tipping path $\frac{1}{2}$ max + $\frac{1}{2}$ second max.

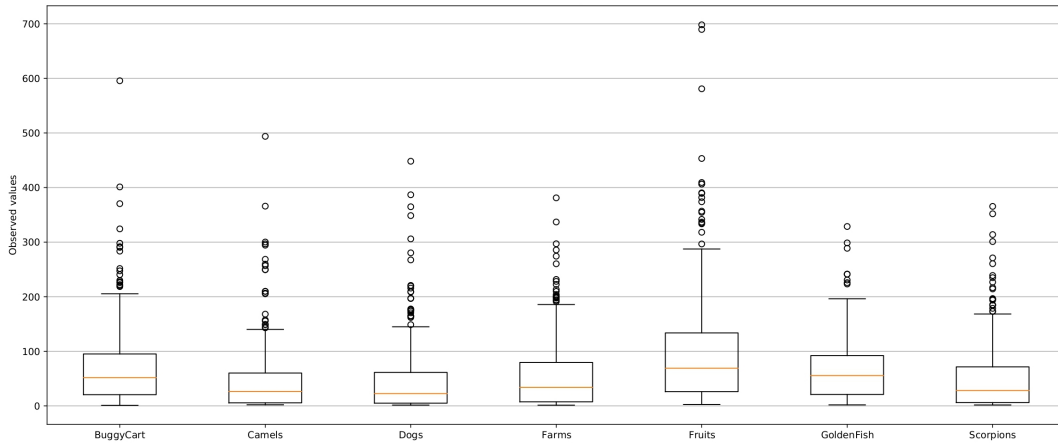


Fig. 1. Fluctuations of the confidence index (12) for the SoftMax path

4.3 Unsupervised feature delineation

In this section, we provide some evidence that our approach can help delineate the features in the images from the training dataset that characterise the network’s decision to assign the image to a specified label. The main novelty about our approach is that this task is performed in a completely unsupervised manner.

Our results are as follows. Figures 7, 8 and 9 contain comparisons between original images with their tipping points for the ResNet50, ResNet101 and ResNet152 respectively. The first row displays a sample of one picture per class,

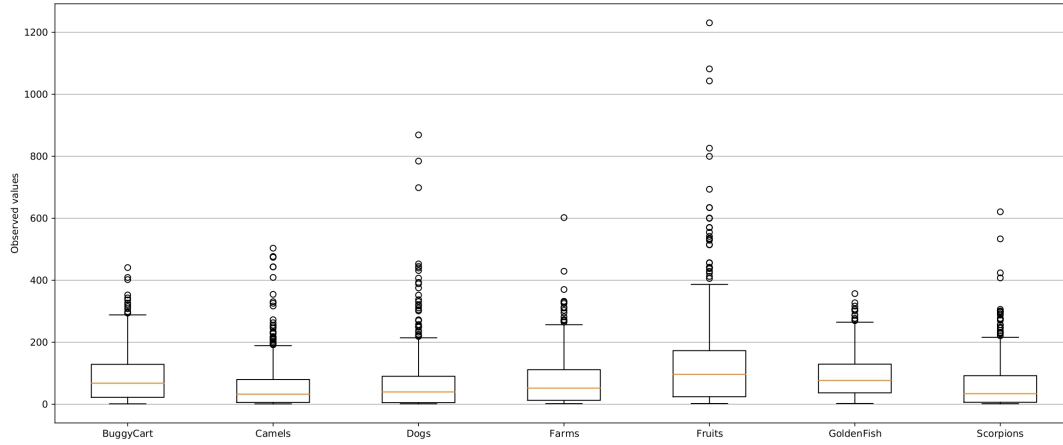


Fig. 2. Fluctuations of the confidence index (12) for the Dirac path

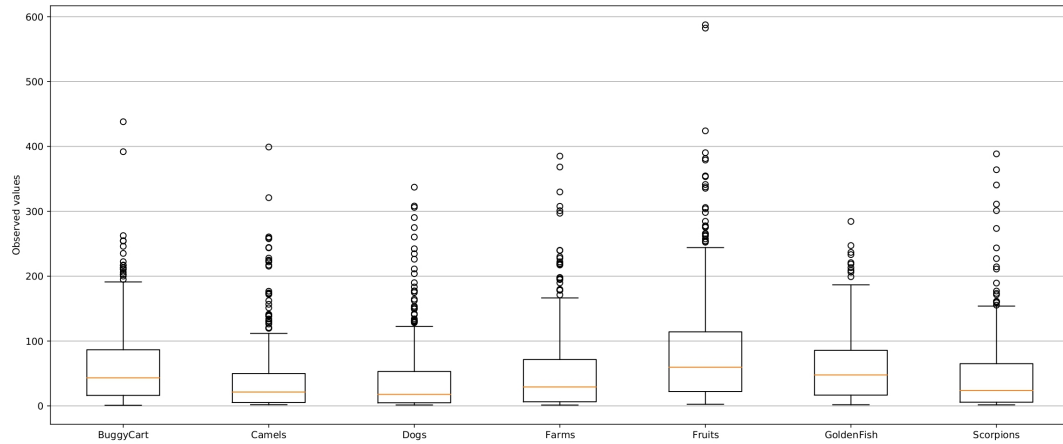


Fig. 3. Fluctuations of the confidence index (12) for the $\frac{1}{2} \frac{1}{2}$ path

while the second row displays the associated tipping point. The third row is the representation of the difference between the original picture and the respective tipping point. Finally the fourth row is a “saturated” rescaled version of the pictures displayed in the third row. The function used for the saturation filter was $x \rightarrow \exp(a \cdot |x|) - 1$ with $a = 1000$ after what the picture was clipped, the negative values were set to 0, and the values greater than 1 set to 1.

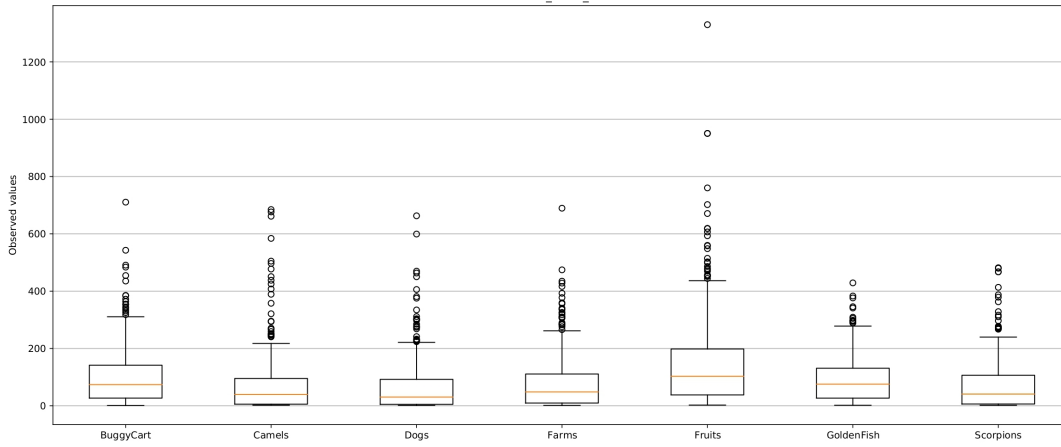


Fig. 4. Fluctuations of the confidence index (12) for the Shifted vector path

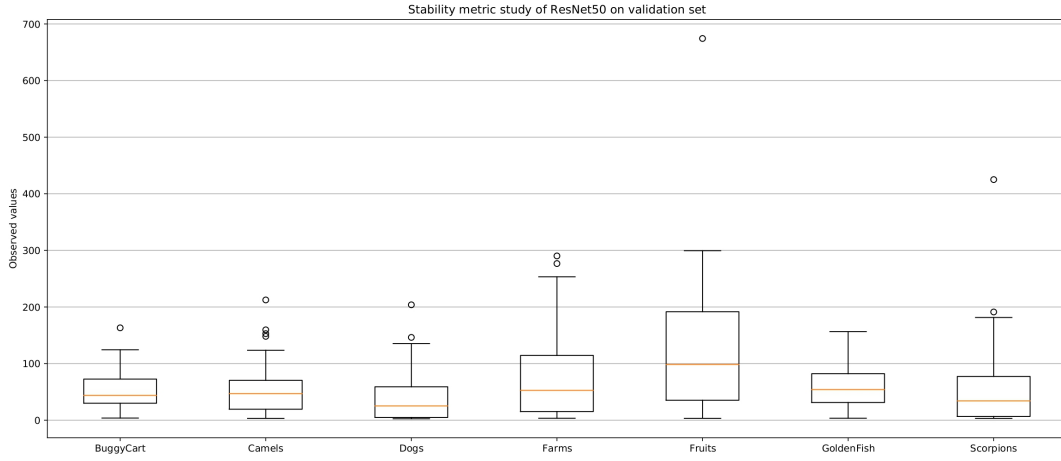


Fig. 5. Fluctuations of the confidence index (12) of ResNet50 on the validation set

4.4 Data augmentation using the tipping points on the active manifold

Our last experiment consists in analysing the performance gain of augmenting the training set with the tipping points associated with the images from the training set, using the fine-tuned ResNet50. We idAugment the training data set with these tipping points (associated with the correct labels) and fine-tune a new ResNet50 that we called ResNet50PAD (PAD for post augmented data). In the

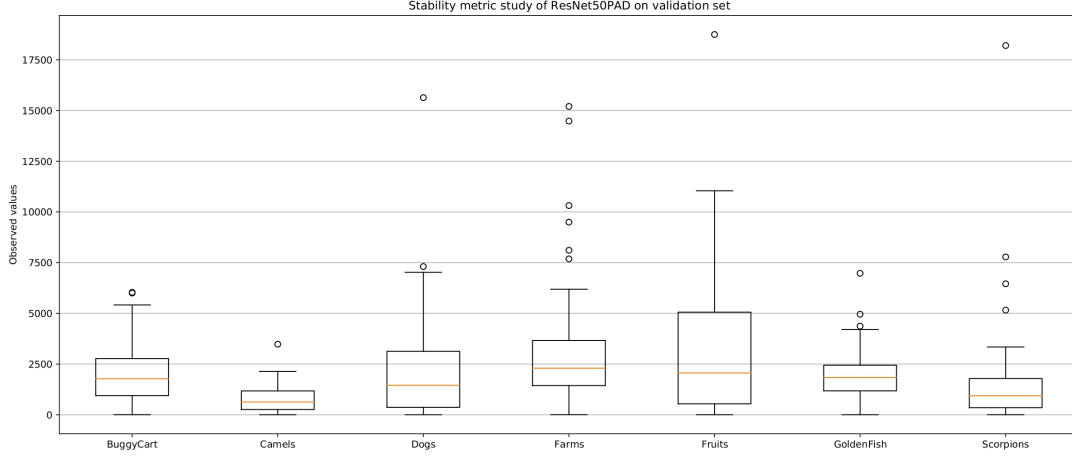


Fig. 6. Fluctuations of the confidence index (12) of ResNet50PAD on the validation set

following table we compare the accuracy of ResNet50 and ResNet50PAD on the original training data set (described at the beginning of the section Numerical experiments) and the validation data set.

Name	Original training set	Validation set
ResNet50	79%	88%
ResNet50PAD	87%	90%

Figures 5, 6 illustrate the behavior of the confidence index on the validation data set of the two considered ResNets. We observed a significant increase in the confidence index. We also observed better accuracy on the training (original training set with 450 pictures per class) and validation set (50 pictures per class). This small experiment already shows that data augmentation studies can benefit from incorporating images for which the confidence index computed using the nonsmooth manifold approach was low.

5 Conclusion

In this paper we propose a new tool for analysing the confidence associated with a neural network’s decision regarding the classification problem for a set of images. Our new tool exploits the concept of nonsmooth active manifolds. Our contributions are threefold:

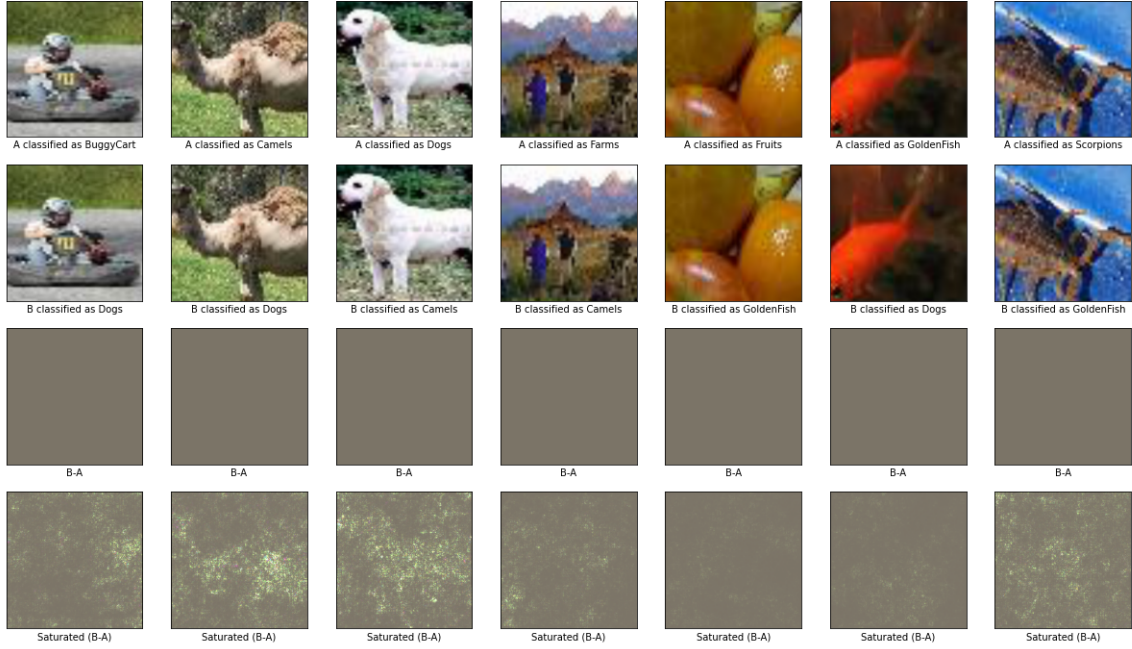


Fig. 7. A: is the input image. B: is the tipping points. B-A: is the difference between A and B. Saturated (B-A) is a saturated version of B-A. All these results are for ResNet50

- first, we have defined an extension to the active manifold theory of [1] that seamlessly adapts to any deep network architecture,
- next, we used our approach to define a new index of confidence that can easily be computed for any image by studying its distance to an associated tipping point in the nonsmooth active manifold and we analysed the performance of different ResNet architectures based on this new index
- finally, we showed that our approach can be also applied to the difficult problems of unsupervised feature extraction (in a weak sense) and principled data augmentation.

In future works we plan to refine the design of our nonsmooth active manifolds by using penalisation based on the Mumford Shah functional in order to improve our unsupervised feature extraction approach by selecting meaningful homogeneous patches. We expect that by improving the feature extraction stage, all the other tasks will also be better addressed.

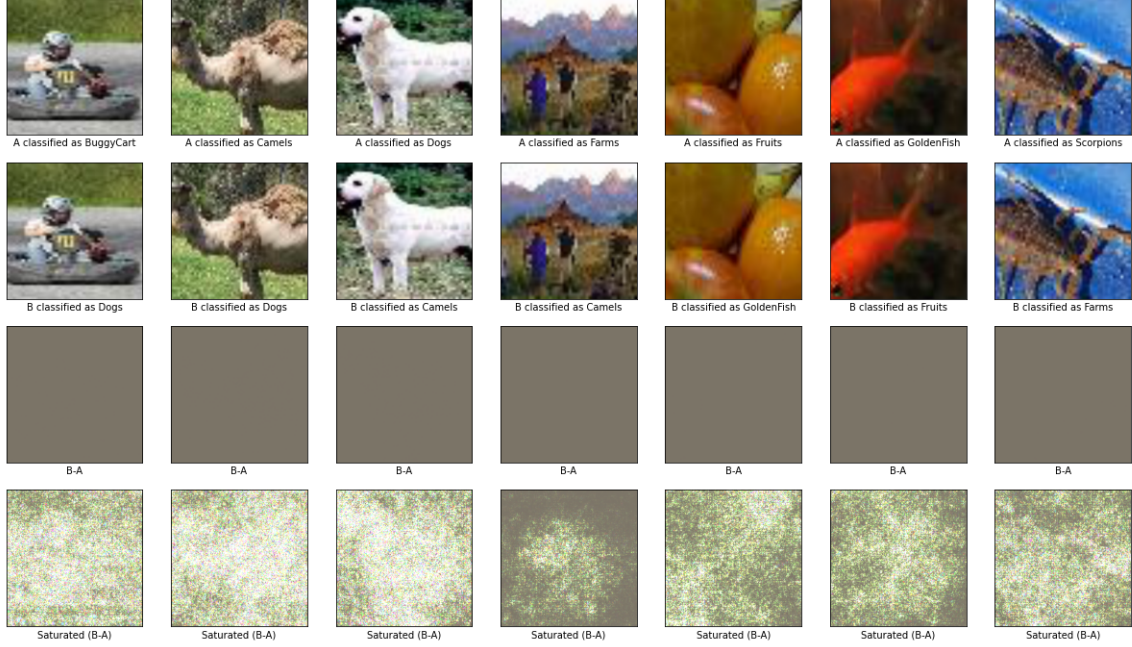


Fig. 8. A: is the input image. B: is the tipping points. B-A: is the difference between A and B. Saturated (B-A) is a saturated version of B-A. All these results are for ResNet101

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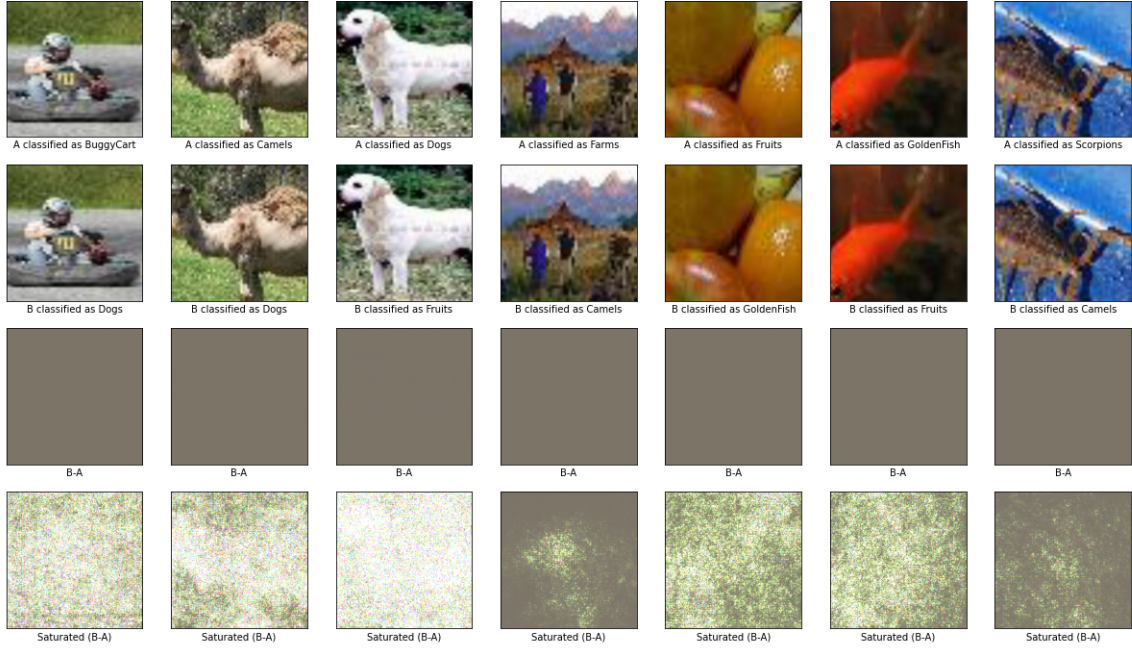


Fig. 9. A: is the input image. B: is the tipping points. B-A: is the difference between A and B. Saturated (B-A) is a saturated version of B-A. All these results are for ResNet152

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